

$^{34}\text{S}(\text{d},\text{p}\gamma)$ 1972Fr11,1975VaYG,1970Bu18

1972Fr11: 3.50- and 4.00-MeV deuteron beams were produced from the 4-MV Van de Graaff accelerator at the Nuclear Research Center of Strasbourg, France. Targets were antimony sulphide enriched to 85.6%, $550\text{ }\mu\text{g}/\text{cm}^2$ on a 0.03 mm molybdenum layer and $150\text{ }\mu\text{g}/\text{cm}^2$ on a 0.05 mm silver layer. Protons were detected using a surface-barrier annular silicon detector at 180° . γ rays were detected using a 40 cm^3 Ge(Li) detector at 0° , 55° and 90° with FWHM=3.2 keV at $E_\gamma=1.332\text{ MeV}$ in the first experiment and a NaI crystal at 0° , 30° , 45° , 60° and 90° in the second experiment. Measured E_p , E_γ , I_γ , and $\text{p}\gamma(\theta)$ -coin. Deduced 12 levels, J, γ -ray branching ratios, and mixing ratios. Deduced $T_{1/2}$ for 7 levels using the Doppler-Shift Attenuation Method (DSAM).

1975VaYG,1972Va07: A 4.72-MeV deuteron beam of 175 nA was provided from the Groningen 5 MV Van de Graaff generator. Targets were $80\text{--}\mu\text{g}/\text{cm}^2$ Zn^{34}S enriched to 89% evaporated onto $10\text{--}\mu\text{g}/\text{cm}^2$ Formva plus $10\text{--}\mu\text{g}/\text{cm}^2$ carbon. Protons were detected using a 2-mm annular silicon detector. γ rays were detected using a $7.6\text{-cm}\times 7.6\text{-cm}$ NaI(Tl) detector. Measured E_p , E_γ , I_γ , and $\text{p}\gamma(\theta)$ -coin. Deduced levels, J, γ -ray branching ratios, and mixing ratios. Also see **1971VaZG**.

1970Bu18: 2-3.25 MeV deuteron beams were produced from the University of Arizona 5.5 MV Van de Graaff accelerator. Targets were $480\text{ }120\text{-}\mu\text{g}/\text{cm}^2$ Ag_2S made by heating 67% enriched ^{34}S powder and silver foil. Protons were detected using a 1-mm annular silicon surface barrier detectors and γ rays were detected using a $7.62\text{ cm}\times 7.62\text{ cm}$ NaI(Tl) detector in the angular correlation experiment and a Ge(Li) detector in the lifetime experiment. Measured E_p , E_γ , I_γ , and $\text{p}\gamma(\theta)$ -coin. Deduced 3 levels, J, γ -ray branching ratios, and mixing ratios. Deduced $T_{1/2}$ for 3 levels using the Doppler-shift attenuation method (DSAM).

2024Co04: A 16-MeV deuteron beam from the John D. Fox Superconducting Linear Accelerator, FSU. Particles were momentum-analyzed by the Super-Enge Split-Pole Spectrograph and identified using a position-sensitive ionization chamber with two anode wires in the isobutane gas (ΔE) and a large plastic scintillator (E) at the focal plane. γ rays were detected using a CeBr_3 Array of five CeBr_3 detectors. Measured $\text{p}\gamma$ delayed coincidence time spectra and deduced $T_{1/2}$ for the 1991 level of ^{35}S .

1971Pr11: A 4.48-MeV deuteron beam was produced from the insulated-core transformer tandem accelerator at the Aerospace Research Laboratories (ARL). The target was Ag_2S enriched to 85.61% in ^{34}S . Protons were detected using a $500\text{ }\mu\text{m}$ -thick, 50-mm^2 silicon detector and γ rays were detected using a 1-in.-thick, 1.5-in. in diameter NE111 scintillator. Measured $\text{p}\gamma$ -delayed coincidence time spectrum and deduced $T_{1/2}$ for the 1.99-MeV level.

1972Go31: A 3.3-MeV deuteron beam was produced from the BNL 3.5-MeV van de Graaff facility. The target was Sb_2S_3 enriched to 86% in ^{34}S . γ rays were detected using a Ge(Li) detector. Measured $E_\gamma=1572.24\text{ }17$ deexciting the first excited state.

 ^{35}S Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}$	Comments
0	$3/2^+$		J^π : from the Adopted Levels.
1572.33 20	$1/2^+$	$>1.3\text{ ps}$	E(level): weighted average of 1574.2 10 (1970Bu18), 1572.3 3 (1972Fr11), and 1572.28 17 (1972Go31). J^π : spin= $1/2$ from isotropic $\gamma(\theta)$ in 1972Fr11 and 1972Va07 . $T_{1/2}$: $>1.3\text{ ps}$ (1970Bu18 , $\tau>1.8\text{ ps}$). $>1.1\text{ ps}$ (1972Fr11 , $\tau>1.6\text{ ps}$).
1992.9 17	$7/2^-$	$1.03\text{ ns } 5$	E(level): unweighted average of 1994.6 10 (1970Bu18) and 1991.2 3 (1972Fr11). J^π : spin= $5/2$, $7/2$ from $\gamma(\theta)$ in 1972Fr11 , 1972Va07 , and 1975VaYG . $T_{1/2}$: Lifetime $\tau = 1.48\text{ ns } 7$ from weighted average of $\tau = 1.7\text{ ns } 3$ (2024Co04) and $\tau = 1.47\text{ ns } 7$ (1971Pr11). Other: $\tau > 4.5\text{ ps}$ using DSAM (1970Bu18).
2348.8 13	$3/2^-$	$0.71\text{ ps } 15$	E(level): unweighted average of 2350.0 10 (1970Bu18) and 2347.5 2 (1972Fr11). J^π : spin from $\gamma(\theta)$ in 1972Va07 , 1975VaYG , and 1972Fr11 . $T_{1/2}$: Lifetime $\tau = 1.03\text{ ns } 21$ from weighted average of $\tau = 0.8\text{ ps } 2$ (1970Bu18) and $\tau = 1.22\text{ ps } 18$ (1972Fr11).
2718.0 10	$5/2^+$	$69\text{ fs } 24$	J^π : spin= $3/2$, $5/2$, $7/2$ from $\gamma(\theta)$ in 1972Va07 and 1972Fr11 . $2717.9\gamma\text{ M1+E2 to } 3/2^+$. $T_{1/2}$: from $\tau=100\text{ fs } 35$ in 1972Fr11 .
2935 3	$3/2^+$		J^π : spin= $3/2$, $5/2$ from $\gamma(\theta)$ in 1972Va07 and 1975VaYG .
3421.0 10	$5/2^+$	$<69\text{ fs}$	J^π : spin= $3/2$, $5/2$, $7/2$ from $\gamma(\theta)$ in 1972Fr11 . $T_{1/2}$: from $\tau<100\text{ fs}$ in 1972Fr11 .
3558	$(3/2^-, 5/2^+)$		E(level): rounded to the nearest integer from the Adopted Levels. 1972Fr11 quotes 3563 8 from 1969Mo12 without reporting their own E(level).
3592 5	$(1/2\text{ to } 7/2)^+$		
3802.0 10	$3/2^-$	$25\text{ fs } 18$	J^π : spin= $3/2$ from $\gamma(\theta)$ in 1972Fr11 . $T_{1/2}$: from $\tau=36\text{ fs } 26$ in 1972Fr11 .
3815 3			

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$^{34}\text{S}(\text{d},\text{p}\gamma)$ **1972Fr11,1975VaYG,1970Bu18 (continued)** ^{35}S Levels (continued)

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}$	Comments
3885.0 10	(3/2 ⁻ ,5/2)		
4025.0 10	(1/2 to 5/2) ⁺		
4107 2	(1/2,3/2,5/2) ⁺	<55 fs	$T_{1/2}$: from $\tau < 80$ fs in 1972Fr11 .
4189 2	1/2 ⁻	<35 fs	$T_{1/2}$: from $\tau < 50$ fs in 1972Fr11 .
4480 2	(1/2,3/2,5/2) ⁺	<62 fs	$T_{1/2}$: from $\tau < 90$ fs in 1972Fr11 .

[†] From the average of available measurements for the lowest three excited states, and from [1972Fr11](#) for higher-lying states.

[‡] From the Adopted Levels. Supporting arguments from this dataset are given under comments if available.

 $\gamma(^{35}\text{S})$

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.	δ	Comments
1572.33	1/2 ⁺	1572.24 17	100	0	3/2 ⁺			E_γ : from 1972Go31 . $A_2 = -0.09$ 5, $A_4 = +0.07$ 6 (1972Fr11); $A_2 = -0.007$ 10, $A_4 = +0.004$ 20 (1975VaYG).
1992.9	7/2 ⁻	1992.8	100	0	3/2 ⁺	M2+E3	+0.17 6	Mult.: D+Q or Q+O from $\gamma(\theta)$; E2+M3 is ruled out by RUL. D+Q is inconsistent with the $J^\pi = 7/2^-$ from the Adopted Levels. δ : unweighted average of $\delta(\text{E3/M2}) = 0.11$ 4 (1970Bu18), +0.11 4 (1972Fr11), and +0.28 3 (1972Va07,1975VaYG) for $J_i = 7/2$. Others: $\delta(\text{Q/D}) = -1.9$ 7 (1970Bu18) and -0.43 3 (1972Va07,1975VaYG) for $J_i = 5/2$. $A_2 = +0.35$ 4, $A_4 = -0.39$ 4 (1972Fr11); $A_2 = +0.16$ 3, $A_4 = -0.10$ 4 (1975VaYG).
2348.8	3/2 ⁻	776.5	27 2	1572.33	1/2 ⁺	(E1)		Mult., δ : D(+Q) from $\gamma(\theta)$, $\Delta\pi = \text{yes}$ from level scheme. M2 component is less likely. $\delta < 0.04$ from $\gamma(\theta)$ and RUL in 1975VaYG , superseding $0 < \delta < +1.7$ from $\gamma(\theta)$ in 1972Va07 . $A_2 = -0.37$ 3, $A_4 = -0.03$ 3 (1972Fr11); $A_2 = -0.38$ 2, $A_4 = +0.01$ 4 (1975VaYG).
		2348.7	73 2	0	3/2 ⁺	(E1)		Mult., δ : D(+Q) from $\gamma(\theta)$, $\Delta\pi = \text{yes}$ from level scheme. M2 component is less likely. $\delta < 0.04$ from $\gamma(\theta)$ and RUL in 1975VaYG , superseding -0.27 12 from $\gamma(\theta)$ in 1972Va07 ; +0.01 4 from $\gamma(\theta)$ in 1972Fr11 assuming pure E1 for 776.5 γ . $A_2 = +0.33$ 2, $A_4 = -0.05$ 2 (1972Fr11); $A_2 = +0.30$ 2, $A_4 = -0.02$ 4 (1975VaYG).
2718.0	5/2 ⁺	369.2 725.1 1145.7 2717.9	<5 <5 <5 100	2348.8 1992.9 1572.33 0	3/2 ⁻ 7/2 ⁻ 1/2 ⁺ 3/2 ⁺	M1+E2	-0.39 5	Mult., δ : D+Q from $\gamma(\theta)$ in 1975VaYG with $-6.9 < \delta < -0.11$ or $+0.42 < \delta < +6.8$ for $J_i = 3/2$, -0.39 5 or -15.0 34 for $J_i = 5/2$, 0.18 4 or 3.8 7 for $J_i = 7/2$. E1+M2 is ruled out by RUL. $\delta = -0.39$ 5 is adopted here based on $J^\pi = 5/2^+$ from the Adopted Levels. -15.0 34 is unlikely based on RUL. $A_2 = +0.47$ 6, $A_4 = -0.08$ 7 (1972Fr11); $A_2 = +0.22$ 2, $A_4 = -0.02$ 3 (1975VaYG).
2935	3/2 ⁺	586	<10	2348.8	3/2 ⁻			

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$^{34}\text{S}(\text{d},\text{p}\gamma)$ **1972Fr11,1975VaYG,1970Bu18** (continued) $\gamma(^{35}\text{S})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Comments
2935	$3/2^+$	942	<20	1992.9	$7/2^-$	
		1363	<15	1572.33	$1/2^+$	
		2935	100	0	$3/2^+$	$A_2=-0.06$ 3, $A_4=-0.01$ 6 (1975VaYG).
3421.0	$5/2^+$	486	<4	2935	$3/2^+$	
		703.0	<6	2718.0	$5/2^+$	
		1072.2	<7	2348.8	$3/2^-$	
		1428.1	<30	1992.9	$7/2^-$	
		1848.6	<7	1572.33	$1/2^+$	
		3420.8	100	0	$3/2^+$	$A_2=-0.54$ 8, $A_4=-0.11$ 7 (1972Fr11).
3592	$(1/2 \text{ to } 7/2)^+$	657	<10	2935	$3/2^+$	
		874	<11	2718.0	$5/2^+$	
		1243	<15	2348.8	$3/2^-$	
		1599	<17	1992.9	$7/2^-$	
		2020	<17	1572.33	$1/2^+$	
		3592	100	0	$3/2^+$	
3802.0	$3/2^-$	867	<4	2935	$3/2^+$	
		1084.0	5 2	2718.0	$5/2^+$	
		1453.2	8 3	2348.8	$3/2^-$	
		1809.1	<4	1992.9	$7/2^-$	
		2229.6	42 4	1572.33	$1/2^+$	$A_2=-0.56$ 15, $A_4=0.0$ 1 (1972Fr11).
		3801.8	45 4	0	$3/2^+$	$A_2=+0.19$ 12, $A_4=0.0$ 1 (1972Fr11).
3815		1822	100	1992.9	$7/2^-$	
3885.0	$(3/2^-, 5/2)$	464.0	<30	3421.0	$5/2^+$	
		950	<12	2935	$3/2^+$	
		1167.0	<15	2718.0	$5/2^+$	
		1536.2	<18	2348.8	$3/2^-$	
		1892.1	100	1992.9	$7/2^-$	
		2312.6	<16	1572.33	$1/2^+$	
		3884.8	<16	0	$3/2^+$	
4025.0	$(1/2 \text{ to } 5/2)^+$	467	<5	3558	$(3/2^-, 5/2^+)$	
		604.0	<10	3421.0	$5/2^+$	
		1090	31 8	2935	$3/2^+$	
		1307.0	<8	2718.0	$5/2^+$	
		1676.2	33 12	2348.8	$3/2^-$	
		2032.0	<10	1992.9	$7/2^-$	
		2452.6	36 12	1572.33	$1/2^+$	
		4024.8	<13	0	$3/2^+$	
4107	$(1/2, 3/2, 5/2)^+$	549	<4	3558	$(3/2^-, 5/2^+)$	
		686	<4	3421.0	$5/2^+$	
		1172	13 6	2935	$3/2^+$	
		1389	<10	2718.0	$5/2^+$	
		1758	<5	2348.8	$3/2^-$	
		2114	<7	1992.9	$7/2^-$	
		2535	<5	1572.33	$1/2^+$	
		4107	87 6	0	$3/2^+$	
4189	$1/2^-$	387	<4	3802.0	$3/2^-$	
		597	<4	3592	$(1/2 \text{ to } 7/2)^+$	
		631	<4	3558	$(3/2^-, 5/2^+)$	
		768	<4	3421.0	$5/2^+$	
		1254	<5	2935	$3/2^+$	
		1471	<5	2718.0	$5/2^+$	
		1840	48 5	2348.8	$3/2^-$	
		2196	<8	1992.9	$7/2^-$	
		2617	<6	1572.33	$1/2^+$	
		4189	52 5	0	$3/2^+$	

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$^{34}\text{S}(\text{d},\text{p}\gamma)$ **1972Fr11,1975VaYG,1970Bu18** (continued) $\gamma(^{35}\text{S})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π
4480	$(1/2,3/2,5/2)^+$	595	<5	3885.0	$(3/2^-,5/2)$
		678	<5	3802.0	$3/2^-$
		888	<5	3592	$(1/2 \text{ to } 7/2)^+$
		922	<5	3558	$(3/2^-,5/2^+)$
		1059	<10	3421.0	$5/2^+$
		1545	<7	2935	$3/2^+$
		1762	61 7	2718.0	$5/2^+$
		2131	<8	2348.8	$3/2^-$
		2487	<10	1992.9	$7/2^-$
		2908	39 7	1572.33	$1/2^+$
		4480	<12	0	$3/2^+$

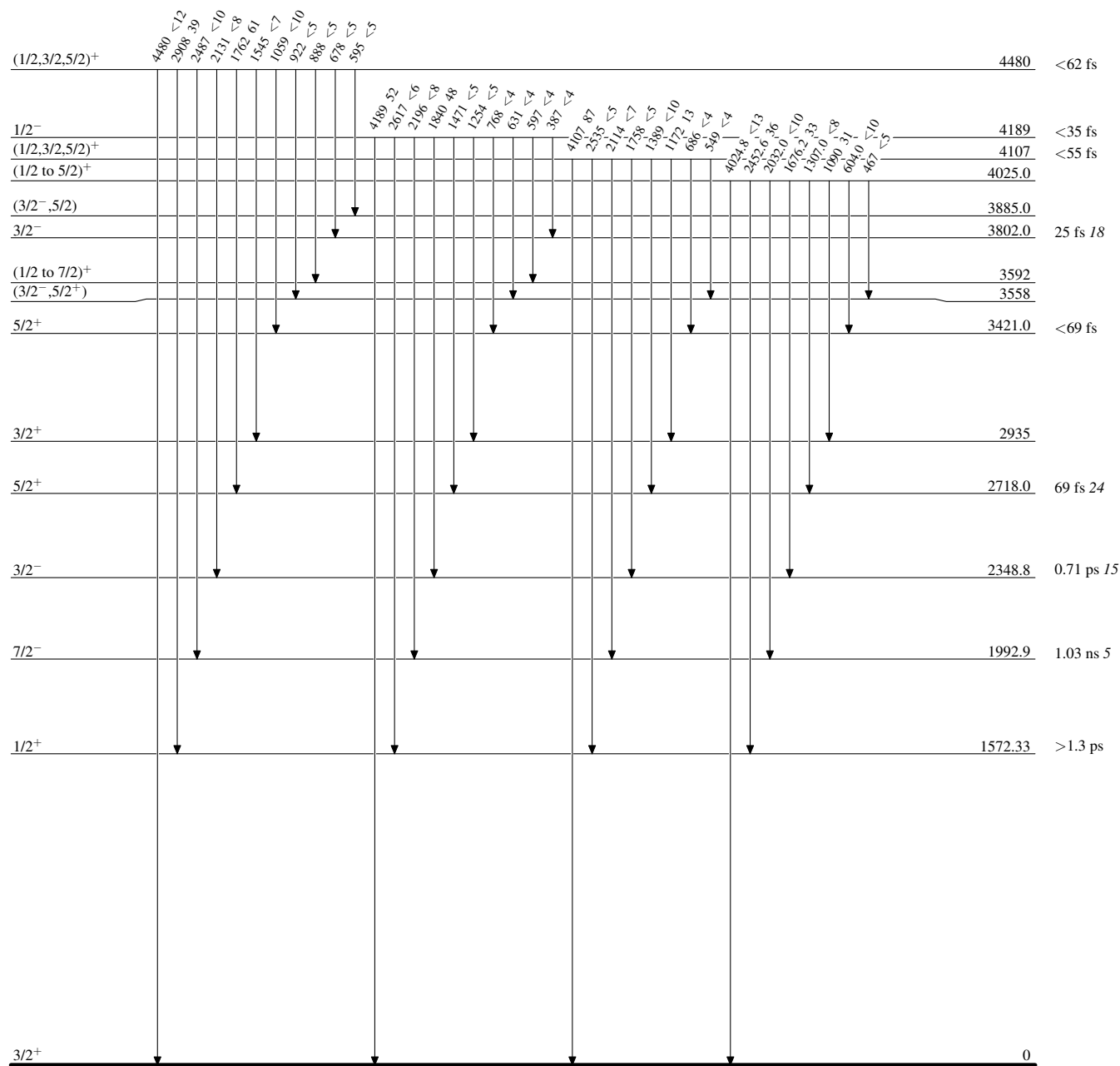
† Deduced by evaluators from level-energy difference, unless otherwise noted.

‡ From **1972Fr11**, unless otherwise noted.

$^{34}\text{S}(\text{d,p}\gamma)$ 1972Fr11,1975VaYG,1970Bu18

Level Scheme

Intensities: % photon branching from each level

 $^{35}_{16}\text{S}_{19}$

